



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

ASSIGNMENT 2

Aadhya Maheshwari- 1024150212

1. Create a structure in C++ containing the details of students as details below and a main function to execute the structure.

CODE:

```
#include <iostream>
using namespace std;

struct Student{
    string name;
    int rollNo;
    string degree;
    string hostel;
    float cgpa;
    void addDetails(){
        cout << "Enter Name: ";
        cin >> name;
        cout << "Enter Roll No: ";
        cin >> rollNo;
        cout << "Enter Degree: ";
        cin >> degree;
        cout << "Enter Hostel: ";
        cin >> hostel;
        cout << "Enter CGPA: ";
        cin >> cgpa;}
    void updateDetails() {
        cout << "Enter new Degree: ";
        cin >> degree;}
    void updateCGPA() {
        cout << "Enter new CGPA: ";
        cin >> cgpa;}
    void updateHostel() {
        cout << "Enter new Hostel: ";
        cin >> hostel;}
    void displayDetails() {
        cout << "\nStudent Details\n";
```

```
        cout << "Name: " << name << endl;
        cout << "Roll No: " << rollNo << endl;
        cout << "Degree: " << degree << endl;
        cout << "Hostel: " << hostel << endl;
        cout << "CGPA: " << cgpa << endl;}};

int main() {
    Student s;
    s.addDetails();
    s.displayDetails();
    s.updateDetails();
    s.updateHostel();
    s.updateCGPA();
    cout << "\nAfter Updating Details: ";
    s.displayDetails();
    return 0;}
```

Enter Name: Aadhya

Enter Roll No: 212

Enter Degree: B.E

Enter Hostel: Q

Enter CGPA: 9

Student Details

Name: Aadhya

Roll No: 212

Degree: B.E

Hostel: Q

CGPA: 9

Enter new Degree: B.Tech|

Enter new Hostel: P.G

Enter new CGPA: 10

After Updating Details:

Student Details

Name: Aadhya

Roll No: 212

Degree: B.Tech

Hostel: P.G

CGPA: 10

=== Code Execution Successful ===

2. Differentiate between private and public access. Perform with class instead of structure with data members private and some member functions in private and some in public scope.

Public- Members are accessible from outside the class well.

Private- Members are accessible only within the class.

CODE:Aadhya

```
#include <iostream>
using namespace std;
class Student {
private:
    string name;
    int rollNo;
    float cgpa;
public:
    void addDetails() {
        cout << "Enter Name: "<<endl;
        cin >> name;
        cout << "Enter Roll No: "<<endl;
        cin >> rollNo;
        cout << "Enter CGPA: "<<endl;
        cin >> cgpa;}

    void displayDetails() {
        cout << "\nStudent Details\n";
        cout << "Name: " << name << endl;
        cout << "Roll No: " << rollNo << endl;
        cout << "CGPA: " << cgpa << endl;}};

int main() {
    Student s;
    s.addDetails();
    s.displayDetails();
    return 0;}
```

Enter Name:

Aadhya

Enter Roll No:

212

Enter CGPA:

10

Student Details

Name: Aadhya

Roll No: 212

CGPA: 10

=== Code Execution Successful ===

3. Calling a private member function inside public member functions.

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class trial{ 4 private: 5 void privateFunction(){ 6 cout << "Private function was called from public function."; 7 } 8 public: 9 void publicFunction(){ 10 privateFunction(); 11 }; 12 int main() { 13 trial obj; 14 obj.publicFunction(); 15 return 0; 16 }</pre>	<pre>Private function was called from public function. === Code Execution Successful ===</pre>

4. Define a class Rectangle with variables width and height of integer type along with the following:

- (a) void getdata() to initialize object values
- (b) void calculatearea() to calculate and display the area

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class Rectangle { 4 int width, height; 5 public: 6 void getdata() { 7 cout << "Enter width: "<<endl; 8 cin >> width; 9 cout << "Enter height: "<<endl; 10 cin >> height; 11 } 12 void calculatearea(){ 13 int area=width*height; 14 cout << "Area of Rectangle = " << area<< 15 endl; 16 }; 17 int main(){ 18 Rectangle r; 19 r.getdata(); 20 r.calculatearea(); 21 return 0; 22 }</pre>	<pre>Enter width: 15 Enter height: 10 Area of Rectangle = 150 === Code Execution Successful ===</pre>

5. Define a class Complex with variables real and imaginary along with the following:

(a) void setComplex(float, float) to initialize object values

(b) void displayComplex() to show the complex number

(c) Pass and return objects to calculate the sum of two complex numbers and display the sum

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class Complex { 4 float real, imag; 5 public: 6 void setComplex(float r, float i) { 7 real = r; 8 imag = i; 9 } 10 void displayComplex() { 11 cout << real << " + " << imag << "i" << endl; 12 } 13 Complex add(Complex c) { 14 Complex temp; 15 temp.real = real + c.real; 16 temp.imag = imag + c.imag; 17 return temp; 18 } 19 } 20 int main() { 21 Complex c1, c2, sum; 22 c1.setComplex(2.5, 3.5); 23 c2.setComplex(1.5, 2.5); 24 sum = c1.add(c2); 25 cout << "Sum of complex numbers: "; 26 sum.displayComplex(); 27 return 0; 28 }</pre>	<pre>Sum of complex numbers: 4 + 6i === Code Execution Successful ===</pre>

6. Explain and implement the use of the scope resolution operator (::) in C++. Demonstrate its use for:

a) Defining a member function outside the class

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class Test { 4 public: 5 void show(); 6 } 7 void Test::show() { 8 cout << "Function defined outside the class" << endl; 9 } 10 int main() { 11 Test t; 12 t.show(); 13 return 0; 14 }</pre>	<pre>Function defined outside the class === Code Execution Successful ===</pre>

b) Accessing global and local variables

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 int x = 10; 4 int main() { 5 int x = 20; 6 cout << "Local x = " << x << endl; 7 cout << "Global x = " << ::x << endl; 8 return 0;}</pre>	<pre>Local x = 20 Global x = 10 === Code Execution Successful ===</pre>

c) Accessing static data members of a class

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class Demo { 4 public: 5 static int x; 6 int Demo::x = 5; 7 int main() { 8 cout << Demo::x << endl; 9 return 0;}</pre>	<pre>5 === Code Execution Successful ===</pre>

d) Accessing inbuilt library members

main.cpp	Output
<pre>1 #include <iostream> 2 int main() { 3 std::cout << "Hello World"; 4 return 0;}</pre>	<pre>Hello World === Code Execution Successful ===</pre>

7. Create a program to implement namespace in C++ and use similar variables and functions defined in different code sections.

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 namespace First { 4 int x = 10; 5 void show() { 6 cout << "Inside First namespace" << endl;}} 7 namespace Second { 8 int x = 20; 9 void show() { 10 cout << "Inside Second namespace" << endl;}} 11 int main() { 12 cout << First::x << endl; 13 First::show(); 14 cout << Second::x << endl; 15 Second::show(); 16 return 0;}</pre>	<pre>10 Inside First namespace 20 Inside Second namespace === Code Execution Successful ===</pre>